

Econ 101A

Section 4 (Solutions)

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In Section 3 the tangency recipe (set MRS equal to the price ratio, solve with the budget) did all the work. That recipe assumes an interior optimum and a well-behaved objective. This section is about the two places it breaks. First, the **second-order condition**: the first-order conditions only find stationary points, and with a constraint you check them with the bordered Hessian. Second, the **tricky cases**: perfect substitutes, perfect complements, quasilinear utility, and non-concave objectives, where the optimum is a corner or a kink and blind tangency gives the wrong answer.

1 The Bordered Hessian, Used in Anger

Take $\max_{x,y} f(x,y)$ subject to $g(x,y) = b$, with Lagrangian $\mathcal{L} = f + \lambda(b - g)$ and a stationary point (x^*, y^*, λ^*) . Form the **bordered Hessian**

$$\bar{H} = \begin{bmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{bmatrix}, \quad \mathcal{L}_{xx} = f_{xx} - \lambda g_{xx}, \text{ etc.},$$

all evaluated at the candidate. With two variables and one constraint the rule is simply

$$\det \bar{H} > 0 \Rightarrow \text{local constrained } \mathbf{max}, \quad \det \bar{H} < 0 \Rightarrow \text{local constrained } \mathbf{min}.$$

The general rule (for reference). With n variables and m constraints, you look at the last $n - m$ leading principal minors of $\bar{H}$: they must alternate in sign, and the sign of the largest (the full determinant) is $(-1)^n$ for a maximum and $(-1)^m$ for a minimum. For $n = 2$, $m = 1$ this collapses to the single-determinant rule above.

When you can skip it. If the objective is concave (a utility that is strictly quasi-concave, i.e. convex-to-the-origin indifference curves) and the constraint set is convex (a straight budget line), the tangency point is automatically the global maximum. Reach for the bordered Hessian only when concavity is not obvious, which is exactly the situation in Exercise 3 below.

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2 When Tangency Fails: Four Cases

1. Perfect substitutes, $u = ax + by$. The $MRS = a/b$ is constant, so the indifference curves are straight lines. Unless they happen to be exactly parallel to the budget line, the best bundle is a **corner**: put all income on the good with the higher “bang per buck” a/p_x vs. b/p_y .

2. Perfect complements (Leontief), $u = \min\{ax, by\}$. The indifference curves are L-shaped, with a **kink** on the ray $ax = by$. There is no derivative at the kink, so there is no tangency. You never buy so much of one good that the min is slack, so the optimum sits on $ax = by$; combine that with the budget to solve.

3. Quasilinear utility, $u = v(x) + y$. Here $MRS = v'(x)$ depends only on x . The interior condition $v'(x) = p_x/p_y$ pins down x^* *independently of income*: once you are interior, every extra dollar goes to the linear good y . At low enough income you hit the corner $y = 0$.

4. Non-concave objective. If the “utility” is convex rather than concave, the tangency point is a **minimum**, not a maximum, and the maximum is a corner. This is the case where the second-order condition earns its keep: the FOC alone would point you to the worst affordable bundle.

3 Exercises

Exercise 1 (perfect complements)

A consumer has $u(x, y) = \min\{x, y\}$ and faces prices $p_x, p_y > 0$ and income m .

- (a) Explain why the tangency condition $MRS = p_x/p_y$ cannot be used here.
- (b) Find the optimal bundle (x^*, y^*) and the indirect utility $v(p_x, p_y, m)$.
- (c) Redo it for $u(x, y) = \min\{x, 2y\}$.

Solution. (a) $u = \min\{x, y\}$ is not differentiable on the kink line $x = y$, so MRS is not defined there; and off the kink one good’s marginal utility is zero, so no interior tangency exists.

(b) Buying more of one good than the other only raises the price paid without raising $\min\{x, y\}$, so at the optimum $x = y$. Call it q : the budget gives $(p_x + p_y)q = m$, hence

$$x^* = y^* = \frac{m}{p_x + p_y}, \quad v = \frac{m}{p_x + p_y}.$$

(c) With $u = \min\{x, 2y\}$ the optimum sits on $x = 2y$. Budget: $p_x(2y) + p_y y = m \Rightarrow y^* = \frac{m}{2p_x + p_y}$,

$$x^* = \frac{2m}{2p_x + p_y}, \text{ and } v = \min\{x^*, 2y^*\} = \frac{2m}{2p_x + p_y}.$$

Exercise 2 (quasilinear utility)

Let $u(x, y) = \ln x + y$ with prices $p_x, p_y > 0$, income m , and $x, y \geq 0$.

- (a) Assuming an interior solution, use the tangency condition to find x^* . Show it does not depend on income.
- (b) Find y^* , and state the income level below which the interior solution breaks down.
- (c) What is the optimal bundle when m is below that threshold?

Solution. (a) $MRS = \frac{u_x}{u_y} = \frac{1/x}{1} = \frac{1}{x}$. Setting $\frac{1}{x} = \frac{p_x}{p_y}$ gives $x^* = \frac{p_y}{p_x}$, which has no m in it: demand for x is independent of income.

(b) From the budget $p_x x^* + p_y y^* = m$: $y^* = \frac{m - p_x x^*}{p_y} = \frac{m - p_y}{p_y} = \frac{m}{p_y} - 1$. This is non-negative only if $m \geq p_y$. So the interior solution requires $m \geq p_y$.

(c) If $m < p_y$, the consumer is at the corner $y^* = 0$ and spends everything on x : $x^* = \frac{m}{p_x}$.

Exercise 3 (a non-concave objective: SOC matters)

Consider the (non-standard) problem

$$\max_{x, y \geq 0} f(x, y) = x^2 + y^2 \quad \text{subject to} \quad p_x x + p_y y = m.$$

- (a) Write the FOCs and find the interior stationary point.
- (b) Compute the bordered Hessian there. Is the stationary point a max or a min?
- (c) Where is the true maximum? Give the optimal bundle and the maximized value.
- (d) In one sentence, what is the lesson?

Solution. (a) $\mathcal{L} = x^2 + y^2 + \lambda(m - p_x x - p_y y)$. FOCs: $2x = \lambda p_x$, $2y = \lambda p_y$, and the budget. Dividing, $x/y = p_x/p_y$; with the budget this gives the interior point $x^s = \frac{m p_x}{p_x^2 + p_y^2}$, $y^s = \frac{m p_y}{p_x^2 + p_y^2}$.

(b) With $g = p_x x + p_y y$ (so $g_x = p_x, g_y = p_y$) and $\mathcal{L}_{xx} = \mathcal{L}_{yy} = 2$, $\mathcal{L}_{xy} = 0$:

$$\bar{H} = \begin{bmatrix} 0 & p_x & p_y \\ p_x & 2 & 0 \\ p_y & 0 & 2 \end{bmatrix}, \quad \det \bar{H} = -2p_x^2 - 2p_y^2 < 0.$$

Since $\det \bar{H} < 0$, the interior stationary point is a constrained **minimum** (geometrically, the point on the budget line closest to the origin).

(c) A convex objective on a line segment is maximized at an endpoint. Compare the corners: $f(\frac{m}{p_x}, 0) = \frac{m^2}{p_x^2}$ versus $f(0, \frac{m}{p_y}) = \frac{m^2}{p_y^2}$. The maximum is at the axis of the *cheaper* good; if $p_x < p_y$,

buy only x : $(\frac{m}{p_x}, 0)$ with value $\frac{m^2}{p_x^2}$ (and symmetrically if $p_y < p_x$).

(d) Lesson: the FOCs find a stationary point, not necessarily a maximum. When the objective is not concave, check the second-order condition and the corners, or you will report the worst bundle as if it were the best.